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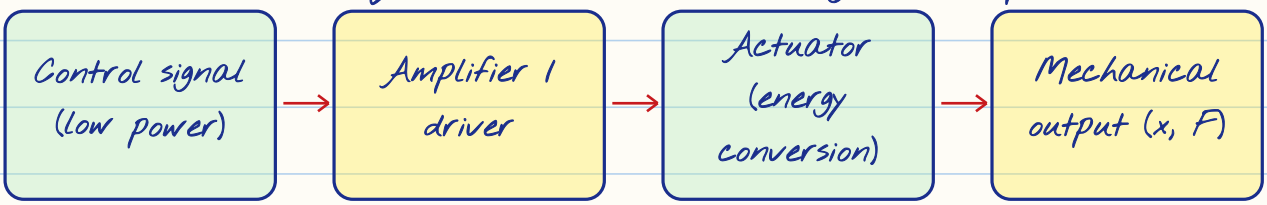
# UNIT-5 : ACTUATORS

IMPORTANT: Syllabus: Introduction, Electrostatic transducers, Electromagnetic transducers, Electrodynamic transducers, Piezoelectric transducers, Electrostrictive transducers, Magnetostrictive transducers, Electrothermal actuators, Comparison of actuation, Applications.

## 5.1 Introduction to Actuators

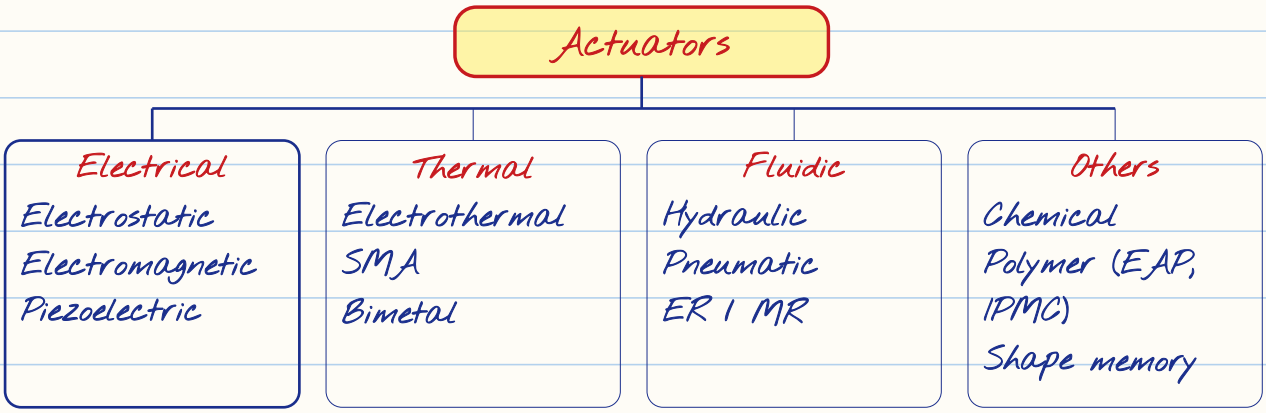
Definition: An actuator is a device that converts electrical (or other) energy into mechanical motion or force to act on a system - it is the "muscle" of a smart system, driven by a control signal from the controller.

- Sensors measure; actuators act . Output: linear displacement, rotation, force, torque, vibration or a change of shape.



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## Classification of actuators




## Important performance parameters

- Stroke (displacement), blocked force, work / energy density (J/m3), power density.
- Bandwidth / frequency response, response time, resolution (nm), hysteresis, linearity.
- Efficiency (output mechanical energy / input energy), driving voltage / current, size and weight, reliability and fatigue life.

Note: A good actuator needs large force + large stroke + high speed + low power, but there is always a trade-off between stroke and force (e.g. piezo = large force, small stroke).

## Transducer = two-way energy conversion

- Many of the actuator mechanisms are reversible: the same device works as an actuator (electrical → mechanical) and as a sensor / generator (mechanical → electrical), e.g. electrostatic, electromagnetic, piezoelectric, magnetostrictive.

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## 5.2 Electrostatic Transducers

**Definition:** Electrostatic actuators use the attractive Coulomb force between oppositely charged conductors (plates) when a voltage is applied across a gap. The energy is stored in the electric field of a capacitor.

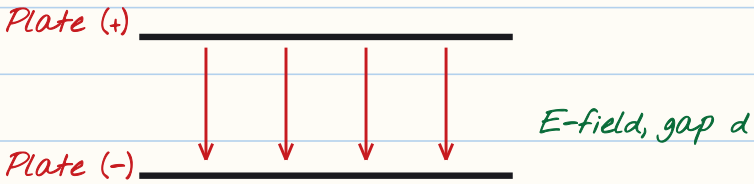



Fig: Parallel plate electrostatic actuator

$$F = \varepsilon_0 A V^2 / (2 d^2)$$ parallel plate force

- $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m, $A$ = plate area, $V$ = applied voltage, $d$ = gap. Force is proportional to $V$ squared and attractive for both polarities.

### Types

- Parallel plate (gap-closing) : large force at small gap; stroke limited. Stable only up to 1/3 of the gap - beyond this the plates snap together ( pull-in instability ; pull-in voltage $V_{PI} = \sqrt{8 K d^3 / 27 \varepsilon_0 A}$).
- Comb drive (lateral) : interdigitated fingers; force independent of displacement : $F = n \varepsilon_0 t V^2 / g$ ($n$ = number of fingers, $t$ = thickness, $g$ = gap); long stroke (10-100 micron), linear.
- Electrostatic motors , scratch-drive actuator , electrostatic membrane / diaphragm (micro-pump, micro-speaker), electret transducers.

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### Features

- Advantages: very simple, low power, fast (KHz-MHz), easy micro-fabrication (surface micromachining, silicon), no magnetic materials, CMOS compatible, low heat, high efficiency at micro scale. Scale favourably as size shrinks.
- Limitations: small force, high voltage (50-100 V), pull-in, stiction, charging / dielectric breakage, humidity sensitivity, small stroke.

Applications: MEMS accelerometers (air bag - ADXL), micro-mirrors (DLP - TI DMD), RF MEMS switches, resonators / filters, micro-grippers, capacitive microphones, optical switches, micro-pumps, inkjet heads, electrostatic chucks, touch panels.

## 5.3 Electromagnetic Transducers

Definition: Electromagnetic actuators produce force by the interaction of a magnetic field and a current or magnetic material. When current flows through a coil it creates a magnetic field that attracts a ferromagnetic armature (reluctance force).

$$F = (B^2 A) / (2\ mu0)$$ force of an electromagnet (a

- B = flux density in the gap, A = pole area, mu0 = 4 pi x $10^{-7}$ H/m. Force is proportional to $(N I)^2 / gap^2$ : N = turns, I = current.

### Types

- Solenoid actuator (plunger / armature in a coil): linear push-pull, high force, short stroke.
- Variable reluctance actuators, relays and contactors, electromagnetic valves.

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- Stepper motors (VR / hybrid), DC motors, brushless DC, linear motors : rotary or linear motion.
- Electromagnetic microactuators (planar coils + permanent magnet, micro-motors, micro relays).

Features

- Advantages: large force, large stroke, simple, wide range of sizes, well developed technology, bi-directional with permanent magnets, low voltage.
- Limitations: heat generation ($I^2R$ losses), bulky at micro scale (force falls as the size$^3$ / poor scaling), magnetic leakage, EMI, hysteresis and non-linearity, needs coil winding.

Applications: Relays, solenoid valves, door locks, fuel injectors, stepper / servo motors, robots, magnetic levitation (maglev), electromagnetic brakes and clutches, MEMS relays, loudspeakers.

## 5.4 Electrodynamic Transducers

Definition: Electrodynamic transducers (moving-coil / voice-coil) produce a force by the Lorentz force on a current-carrying conductor placed in a permanent magnetic field. The reverse (generator) action is Faraday's law - a coil moving in a field induces an emf.

| | |
|---|---|
| $F = B\,L\,I$ | Lorentz force (actuator) |
| $e = B\,L\,v$ | induced emf (generator / sensor) |

- $B$ = flux density (T), $L$ = length of conductor in field (m), $I$ = current (A), $v$ = coil velocity.

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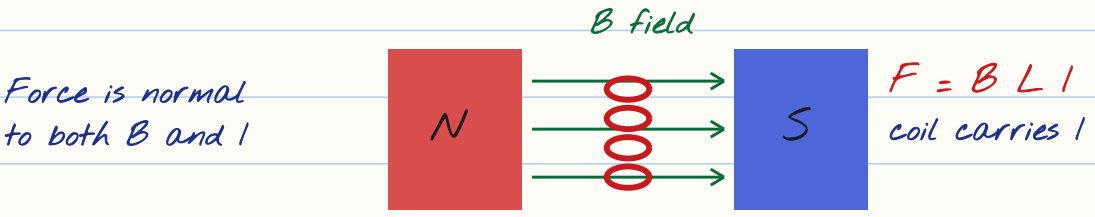



Fig: Voice-coil (electrodynamic) actuator

## Construction (voice coil actuator)

- A cylindrical permanent magnet (NdFeB) + iron pole piece giving a radial field in an air gap; a light coil bobbin moves linearly in the gap, guided by springs / diaphragm.

## Characteristics

- Linear force vs current (no hysteresis), force independent of position over stroke, very fast response (KHz), smooth, accurate positioning, bidirectional (reverse current reverses force), high resolution, low friction.
- Disadvantages: continuous power to hold position (heat), needs magnet and heavy assembly, limited force, magnetic interference.
- Electromagnetic vs electrodynamic : in electromagnetic, force depends on the square of current and is one-directional (attractive) ; in electrodynamic force is linear in current and bidirectional .

Applications: Loudspeakers and headphones , hard-disk drive head actuators (HDD) , camera auto-focus and image stabilisation, shakers (vibration testing), active suspension, microphones (dynamic), seismometers, medical pumps, haptic devices.

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## 5.5 Piezoelectric Transducers

**Definition:** Piezoelectric actuators use the converse piezoelectric effect - when a voltage is applied to a piezoelectric ceramic (PZT), it expands / contracts with high force and sub-nanometre resolution.

$$\text{delta } L = d33 \times V$$ free stroke of a single layer (s

- Strain $S = d\,E$; for PZT $d33 \sim 300\text{-}600$ pC/m (m/V $\times 10^{-12}$), so a 100 micron-thick disc at 100 V gives only about 40 nm; a stack of 100 layers gives ~ 4 micron (0.1% strain).

### Configurations

- **Stack (multilayer) actuator** : many thin ceramic layers (50-200 micron) stacked mechanically in series, electrically in parallel → large force (KN), small stroke (10-200 micron), fast (KHz), low voltage (100-150 V).
- **Bender: Unimorph and Bimorph** (bending): two layers (or one piezo + one passive) - one expands and the other contracts → bends; large displacement (mm), low force.

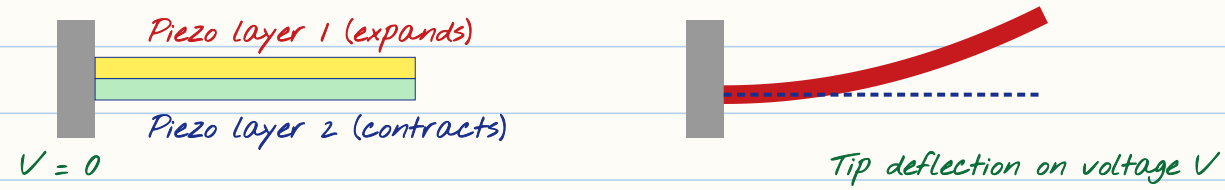



Fig: Piezoelectric bimorph (bending) actuator

- **Amplified actuators** : lever / flexure hinge amplifier, Moonie, Cymbal (end caps), RAINBOW, Thunder (pre-stressed benders), hydraulic amplification.
- Tubular / ring actuators (radial deformation), patch actuators (MFC) (macro fibre composite) for structures.
- **Piezoelectric motors** : ultrasonic travelling-wave motor (USM),

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inchworm , stick-slip, inertial motors - unlimited stroke by repeated steps.

Characteristics

- Advantages: nanometre resolution , fast response (microseconds), high force and stiffness, high energy density, no magnetic field, no gear / backlash, works in vacuum / cryogenic, no power to hold position , compact.
- Limitations: small strain (0.1%), hysteresis (10–15%) and creep, high voltage amplifier, brittle ceramics (cannot take tension), depolarisation at high temp and field, capacitive load (needs current to drive at high freq).

Applications: Precision positioning (AFM / STM, wafer steppers), fuel injectors , inkjet printers , ultrasonic motors in camera lenses, active vibration and noise control, adaptive optics , micro-pumps, ultrasonic cleaning / welding / surgical knives, piezo buzzers, atomisers, MEMS (AlN, PZT films), valves.

## 5.6 Electrostrictive Transducers

Definition: Electrostrictive actuators use the electrostrictive effect of relaxor ceramics such as PMN-PT / PMN-PT-BT where strain is proportional to the square of the electric field ($S = M E^2$); the effect is the same in both field directions.

- Typically a stack of PMN (lead magnesium niobate) multilayers, operated with a DC bias to linearise the response.
- Strain ~ 0.1% at 1–2 kV/mm; stroke 10–20 micron per stack; blocking force high; response in microseconds.

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### Features

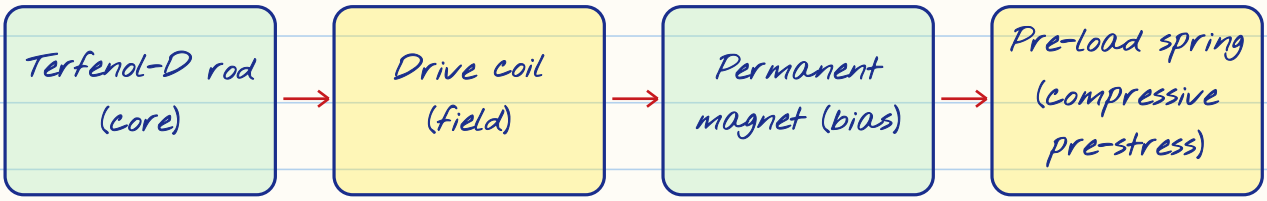
- Advantages: very low hysteresis (< 2%) and no creep , no poling / depoling , high repeatability and stability, excellent positioning accuracy , high energy density, good for dc / quasi-static.
- Limitations: temperature sensitivity (strong variation of strain near the Curie region), non-linear (quadratic), lower strain per volt than PZT at low field, more expensive, not usable as a sensor (no direct effect unless biased), cannot reverse direction of motion, thermal drift and capacitance.

Applications: Deformable mirrors for adaptive optics and telescopes , precision positioners and micro-positioning stages, laser / optical alignment, nano-metrology, vibration isolation, inKjet printer heads, ultrasonic imaging (PMN-PT crystals), SONAR.

## 5.7 Magnetostrictive Transducers

Definition: Magnetostrictive actuators use the Joule effect : a ferromagnetic rod (usually Terfenol-D ) elongates when a magnetic field is applied by a coil ; the inverse (Villari) effect gives sensing.

### Construction




- Strain = lambda up to ~ 1500-2000 ppm (rod of 100 mm gives ~ 150-200 micron); blocking force up to several KN, energy density 14-25 KJ/m3.
- Pre-stress 7-10 MPa aligns the domains perpendicular to the rod

axis, giving more strain on application of field; bias field gives linear, bipolar operation (otherwise strain ~ $H^2$ frequency doubling).

- Laminated rods / powder composites reduce eddy current loss at high frequency.

Features

- Advantages: large strain and force, non-contact magnetic drive (no electrical contact), low voltage (current driven), high Curie temperature (380 C), fast (up to 10-50 KHz), works at cryogenic temperature, high power density, no depolarisation.
- Limitations: costly (Tb, Dy rare earths), brittle, heavy, needs coil and bias magnet (large size), hysteresis (5-10%), eddy-current and magnetic losses at high frequency, temperature sensitivity, low efficiency, flux leakage.

Applications: Sonar and underwater acoustic projectors, ultrasonic cleaners and welders, active vibration / noise control, high-force linear micro-positioners, fuel injectors (diesel), magnetostrictive motors (inchworm), pumps, energy harvesting, active damping of machine tools, valve actuators, medical / dental ultrasonic scalers.

## 5.8 Electrothermal Actuators

Definition: Electrothermal (thermal) actuators produce motion from thermal expansion of a material heated by Joule (resistive) heating, $I^2R$, or an external heat source.

Displacement = alpha x L x delta T.

$$\Delta L = \alpha \times L \times \Delta T$$ thermal expansion

- alpha = coefficient of thermal expansion, L = length, delta T = temperature rise.

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## Types

- **Bimorph (bimetal) actuator** : two layers with different alpha (e.g. Al on Si, Ni on Si, polymer on metal) bonded together; on heating the layer with higher alpha expands more → **bending** (thermostats, micro-mirrors).
- **Hot-and-cold arm (V-shaped, Guckel) actuator** : a thin (hot) arm and a wide (cold) arm in series; the hot arm has higher resistance → heats more → expands more → **lateral deflection** (no stiction, large force, low voltage 3–10 V).

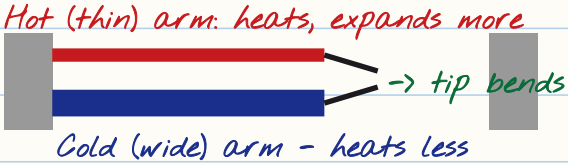



Fig: Electrothermal (hot-arm / V-shaped) actuator

- **Chevron (V-shaped, bent-beam) actuator**: pre-bent beams push the apex when heated; **large force and displacement** ; excellent for micro-gripper.
- **Shape memory alloy actuators** (NiTi wire / film): current heats alloy above Af → contracts (see Unit 2).
- **Thermopneumatic actuators** : heating a fluid in a cavity expands / vaporises it → membrane deflects (micro-pump, valves).
- **Bubble / thermal inkjet** (vapour bubble), **paraffin wax** actuators (large expansion at melting, auto thermostats).

## Features

- Advantages: **very large force** and displacement per area, **low voltage** , simple structure, easy to fabricate (surface micromachining), robust, compatible with CMOS, no EMI.
- Limitations: **slow** (cooling limited: 1–100 Hz), **high power**

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consumption, low efficiency (< 1-10%), temperature effects on the surroundings, ageing and fatigue, difficult to control, heat loss to substrate.

Applications: Micro-grippers, micro-valves and pumps, thermal inkjet printers, thermostats and fire dampers, optical switches / shutters, micro-relays, MEMS positioners, SMA couplings, thermal micro-mirrors, automotive thermostats.

## 5.9 Comparison of Actuation Mechanisms

| Type | Force | Stroke | Speed / Bandwidth | Efficiency | Remarks |
|---|---|---|---|---|---|
| Electrostatic | Low | Small (10-100 micron) | Fast (KHz-MHz) | High (micro) | Needs high V; CMOS compatible |
| Electromagnetic | High | Large | Medium | Medium | Heat, bulky, poor scaling |
| Electrodynamic | Medium | Large | Fast (KHz) | Medium | Linear, bidirectional |
| Piezoelectric | High (KN) | Small (0.1%) | Very fast (KHz-MHz) | High (>50%) | Nano precision, hysteresis |
| Electrostrictive | High | Small (0.1%) | Fast | High | Low hysteresis, temp. sensitive |
| Magnetostrictive | High | Small-medium (0.2%) | Fast (KHz) | Low-medium | Large power density, costly |
| Electrothermal | High | Medium | Slow (Hz) | Low | Low voltage, high power |
| SMA | Very high | Large (4-8%) | Slow (1-5 Hz) | Very low (<5%) | High work density |

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| ER / MR fluid | Controlled damping | - | ms | - | Semi-active clutch / damper |

## Selection criteria

- Stroke needed, force, speed, precision (resolution), power / voltage available, size and weight, cost, environment (temperature, vacuum, EMI), reliability and life.
  - Nano-precision → piezo / electrostrictive; large stroke / force → electromagnetic; micro scale → electrostatic / thermal; high force sonar → magnetostrictive.

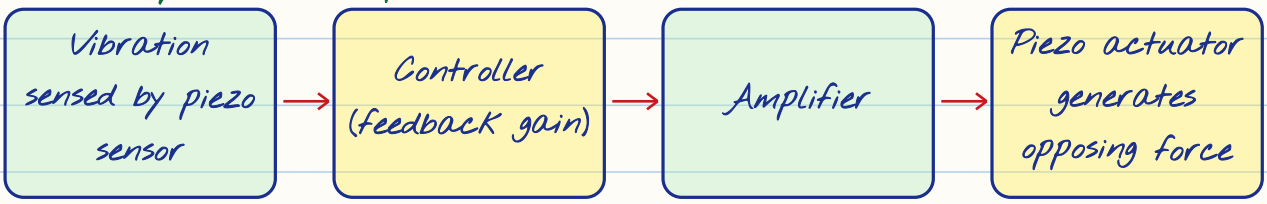
## 5.10 Applications of Actuators

| Field | Actuator used | Application |
|---|---|---|
| Aerospace | Piezo patches, SMA, MR | Active vibration / flutter control, morphing wing, flap control, satellite pointing |
| Automotive | Piezo, electromagnetic, MR | Common-rail fuel injectors, ABS valves, MR suspension, power windows |
| Biomedical | SMA, piezo, electrostatic, thermal | Stents, micro-pumps for insulin, surgical tools, micro-grippers, ultrasound probes |
| Industrial / Robotics | Electrodynamic, electromagnetic, piezo | Precision positioning, grippers, valves, motors, machine-tool damping |
| Consumer electronics | Voice coil, piezo, electrostatic | HDD heads, speakers, camera auto-focus, haptic feedback, inkjet, DLP projector mirrors |
| Optics | Piezo, electrostrictive | Adaptive optics deformable mirrors, fibre alignment, optical switch |

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|---|---|---|
| Civil engineering | MR dampers, piezo, SMA | Earthquake damping, active structural control, smart braces |
| Energy / defence | Magnetostrictive, piezo | SONAR, energy harvesters, noise control, vibration isolation |

### Smart system example - **Active vibration control**




- Result: vibration amplitude falls, structure damping increases; used in aircraft panels, space structures, machine tools.

### Future trends

- MEMS / NEMS actuators, artificial muscles (EAP, CNT yarns), soft robotics, 4D printed actuators, energy-efficient lead-free piezo (KNN, BNT), magnetoelectric and multi-functional actuators, integration with AI control.

## 5.11 Quick Revision / Important Questions

- 1. What is an actuator? Explain the classification and performance parameters.
- 2. Explain the working principle of electrostatic actuators. Derive the force and discuss pull-in.
- 3. Differentiate electromagnetic and electrodynamic transducers with equations.
- 4. Explain piezoelectric actuators: stack, bimorph, ultrasonic motor.
- 5. Write notes on electrostrictive and magnetostrictive actuators.

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- 6. Explain electrothermal actuators with the hot-and-cold arm structure.
- 7. Compare various actuation methods in a table, applications of each.

IMPORTANT: Exam tip: remember the formulas - Electrostatic F = eps0 A V² / 2 d², Lorentz F = B L I, Piezo delta L = d33 V, Thermal delta L = alpha L delta T. Add one application for each and the comparison table.